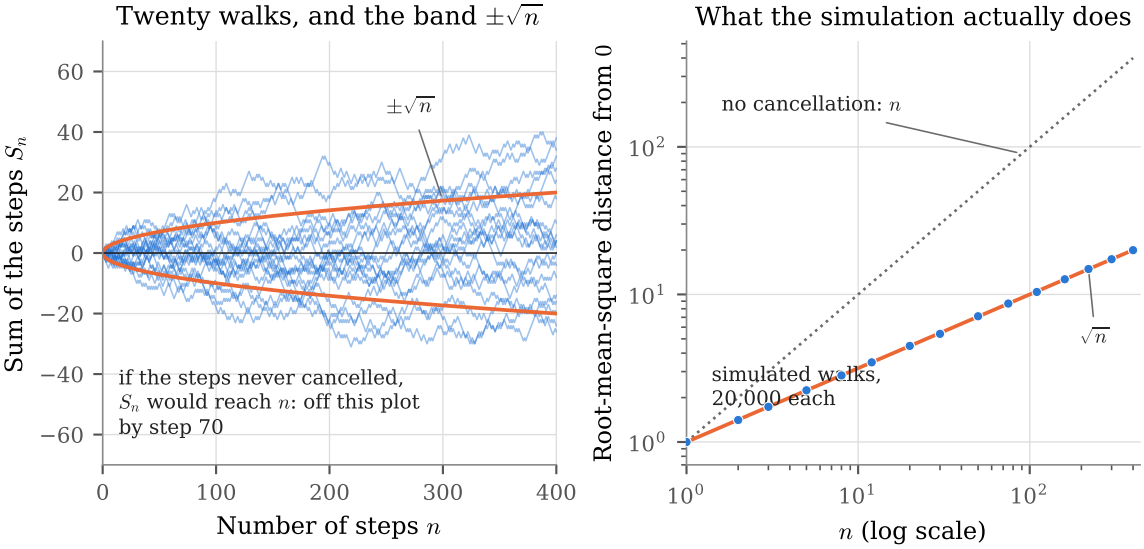
Twenty walks, and the band $\pm\sqrt{n}$
Sum of the steps S_n
Number of steps n
$\pm\sqrt{n}$
if the steps never cancelled,
S_n would reach n: off this plot
by step 70
What the simulation actually does
Root-mean-square distance from 0
n (log scale)
no cancellation: n
$\sqrt{n}$
simulated walks,
20,000 each